

# Generation Interconnection System Impact Study Report

## For Queue Project Q15-0437

Lawrenceville 138 kV

December 24, 2015

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0437. The Interconnection Customer proposes a 196.4 MW wind facility interconnecting at the Lawrenceville 138 kV substation in the Virginia area. The request was received on February 17, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 272 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,397,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

The Skaggs Island - Lawrenceville 138.0 kV is overloaded to 170.7% of its 237 MVA emergency rating for the loss of the N Tower Tap - McCartney 345.0 kV. The project contributes 46.0% DFAX (90.4 MW) to this overload.

Dpc Elroy - N Tower Tap 138.0 kV: 170.6% loading (286 MVA) following the outage of the McCartney - N Tower Tap 345.0 kV; DFAX = 36.7%.

Garden - Dpc Elroy 138.0 kV line: 170.6% loading (223 MVA) following the outage of the N Tower Tap - McCartney 345.0 kV; DFAX = 46.0%.

Skaggs Island - Garden 138.0 kV L: 169.9% loading (230 MVA) following the outage of the McCartney - North Tower Tap 345.0 kV; DFAX = 46.0%.

Bells Mill - Beverly Farms 138.0 kV L: 169.3% loading (212 MVA) following the outage of the Tiger Bay - Dranesville 345.0 kV L; DFAX = 5.4%.

For the loss of Haggin - N Tower Tap 345.0 kV, the Unknown Substation - Walthall 138.0 kV loads to 168.0% of its 317 MVA rating. Project DFAX: 19.1%.

N Tower Tap - Unknown 138.0 kV: 159.4% loading (344 MVA) following the outage of the N Tower Tap - Haggin 345.0 kV; DFAX = 19.1%.

The North Pole - Dixleta Substation 138.0 kV is overloaded to 152.5% of its 465 MVA emergency rating for the loss of the N Pole - Dixleta 138.0 kV Line 2. The project contributes 5.7% DFAX (11.3 MW) to this overload.

For the loss of Rustic - Dixleta 345.0 kV, the Hughson - Dixleta 345.0 kV loads to 152.4% of its 1,599 MVA rating. Project DFAX: 19.8%.

The Bear Island Substation - Chancellor Row 345.0 kV is overloaded to 150.2% of its 1,105 MVA emergency rating for the loss of the Leavitt - Birchwood Substation 345.0 kV line. The project contributes 24.8% DFAX (48.7 MW) to this overload.

Dixleta - N Pole 138.0 kV: 149.9% loading (472 MVA) following the outage of the N Pole - Dixleta 138.0 kV Ckt 1; DFAX = 5.7%.

Drum - Montpelier 138.0 kV: 149.9% loading (432 MVA) following the outage of the Taber - Dixleta 345.0 kV; DFAX = 5.3%.

The Rustic - Dixleta 345.0 kV is overloaded to 141.9% of its 1,612 MVA emergency rating for the loss of the Dixleta - Hughson 345.0 kV. The project contributes 18.0% DFAX (35.3 MW) to this overload.

For the loss of Glenannan - Taber 345.0 kV, the N Pole - Hih Ridge 138.0 kV loads to 141.7% of its 799 MVA rating. Project DFAX: 8.4%.

The Jerome - High Ridge 138.0 kV L is overloaded to 140.9% of its 830 MVA emergency rating for the loss of the Taber Substation - Glenannan 345.0 kV. The project contributes 8.4% DFAX (16.4 MW) to this overload.

For the loss of Mitchell Dam - Haggin 345.0 kV, the Syl Laskin - Elko 138.0 kV loads to 139.7% of its 200 MVA rating. Project DFAX: 6.5%.

For the loss of Taber - Glenannan 345.0 kV, the Fairland - Sunbright 138.0 kV L loads to 136.8% of its 858 MVA rating. Project DFAX: 7.2%.

Fritch - Big Hanaford 138.0 kV: 135.7% loading (281 MVA) following the outage of the Mitchell Dam - Skaggs Island Jct 2 345.0 kV; DFAX = 10.5%.

Birchwood - Dist 1000 138.0 kV: 134.0% loading (304 MVA) following the outage of the Birchwood Substation - Leavitt 345.0 kV; DFAX = 5.3%.

Taber - Glenannan 345.0 kV: 133.4% loading (2,247 MVA) following the outage of the Sierra Pacific Lumber 138-1 kV Auto; DFAX = 20.0%.

For the loss of Glenannan - Taber 345.0 kV Line, the Sunbright - Colesville 138.0 kV loads to 133.3% of its 922 MVA rating. Project DFAX: 7.2%.

Taber 345/138kV XFMR: 131.8% loading (918 MVA) following the outage of the Taber - Dixleta 345.0 kV; DFAX = 5.3%.

The Dixleta - Drum 138.0 kV Line is overloaded to 130.7% of its 799 MVA emergency rating for the loss of the Taber - Glenannan 345.0 kV. The project contributes 5.9% DFAX (11.6 MW) to this overload.

Westport - Wheelabrator Baltimore Refuse 138.0 kV L: 130.3% loading (589 MVA) following the outage of the Taber - Dixleta 345.0 kV; DFAX = 5.8%.

The Taber - Dixleta 345.0 kV is overloaded to 130.1% of its 2,150 MVA emergency rating for the loss of the Sierra Pacific Lumber 138-1 kV Autotransformer. The project contributes 18.0% DFAX (35.3 MW) to this overload.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Lawrenceville - Skaggs Island Substation 138.0 kV L | Rebuild line with higher-capacity conductor | \$4,488,000   | 66% (\$2,953,000)     |
| Dpc Elroy - N Tower Tap 138.0 kV                    | Rebuild line with higher-capacity conductor | \$5,708,000   | 97% (\$5,555,000)     |
| Garden - Dpc Elroy 138.0 kV line                    | Rebuild line with higher-capacity conductor | \$16,708,000  | 66% (\$10,994,000)    |
| Skaggs Island - Garden 138.0 kV L                   | Rebuild line with higher-capacity conductor | \$30,661,000  | 66% (\$20,175,000)    |
| Bells Mill - Beerly Farms 138.0 kV                  | Rebuild line with higher-capacity conductor | \$2,812,000   | 3% (\$80,000)         |
| Unknown Substation - Walthall 138.0 kV              | Rebuild line with higher-capacity conductor | \$15,497,000  | 13% (\$1,953,000)     |

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| N Tower Tap - Unknown 138.0 kV                      | Rebuild line with higher-capacity conductor | \$6,184,000   | 13% (\$779,000)       |
| North Pole - Dixleta Substation 138.0 kV            | Rebuild line with higher-capacity conductor | \$1,080,000   | 2% (\$23,000)         |
| Hughson - Dixleta 345.0 kV                          | Rebuild line with higher-capacity conductor | \$33,036,000  | 2% (\$651,000)        |
| Chancellor Row - Bear Island 345.0 kV               | Rebuild line with higher-capacity conductor | \$37,686,000  | 3% (\$1,093,000)      |
| N Pole - Dixleta 138.0 kV                           | Rebuild line with higher-capacity conductor | \$1,172,000   | 2% (\$25,000)         |
| Drum - Montpelier 138.0 kV                          | Rebuild line with higher-capacity conductor | \$8,493,000   | 3% (\$239,000)        |
| Rustic - Dixleta 345.0 kV                           | Rebuild line with higher-capacity conductor | \$75,997,000  | 2% (\$1,496,000)      |
| North Pole - High Ridge 138.0 kV line               | Rebuild line with higher-capacity conductor | \$2,955,000   | 2% (\$61,000)         |
| Jerome - High Ridge 138.0 kV L                      | Rebuild line with higher-capacity conductor | \$1,555,000   | 2% (\$32,000)         |
| Syl Laskin - Elko 138.0 kV                          | Rebuild line with higher-capacity conductor | \$9,543,000   | 4% (\$386,000)        |
| Subright - Fairland 138.0 kV Line                   | Rebuild line with higher-capacity conductor | \$499,000     | 2% (\$10,000)         |
| Fritch - Big Hanaford 138.0 kV                      | Rebuild line with higher-capacity conductor | \$7,721,000   | 4% (\$349,000)        |
| Birchwood - Dist 1000 138.0 kV                      | Rebuild line with higher-capacity conductor | \$13,829,000  | 8% (\$1,164,000)      |
| Glenannan Substation - Taber 345.0 kV               | Rebuild line with higher-capacity conductor | \$28,325,000  | 2% (\$558,000)        |
| Sunbright - Colesville 138.0 kV                     | Rebuild line with higher-capacity conductor | \$1,106,000   | 2% (\$23,000)         |
| Taber 345 kV/138 kV XFMR                            | Replace transformer with higher-rated unit  | \$13,051,000  | 3% (\$366,000)        |
| Dixleta - Drum 138.0 kV Line                        | Rebuild line with higher-capacity conductor | \$2,470,000   | 2% (\$52,000)         |
| Westport - Wheelabrator Baltimore Refuse 138.0 kV L | Rebuild line with higher-capacity conductor | \$470,000     | 2% (\$11,000)         |
| Taber - Dixleta 345.0 kV                            | Rebuild line with higher-capacity conductor | \$18,730,000  | 2% (\$369,000)        |
|                                                     | Total allocated to project                  |               | \$49,397,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.